UCS320: Introduction to Sustainable Green Computing

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Course Objective: This course aims to provide the fundamental concepts and motivations behind sustainable computing, including the environmental impact of IT systems.

Introduction: Principles of green computing; Environmental impact of computing technologies; Green IT procurement and lifecycle management.

Energy-Aware Software Design: Low-power software design principles; Profiling and optimizing software for energy efficiency; Energy-efficient algorithms and data structures; Mobile computing and battery optimization.

Sustainable Data Centers and Cloud Computing: Overview of data centre infrastructure and energy consumption; Green data centres design, cooling, and energy management; Virtualization and server consolidation; Renewable energy in powering data centres; Carbon footprint measurement and reduction in cloud computing.

Course Outcomes (COs)

On completion of the course the students will be able to:

1. Recall the design principles for low-power software.

2. Apply optimization techniques while designing software.

3. Implement energy efficient data structures for developing algorithms.

4. Analyze the energy usage and carbon footprint of hardware, software, and network infrastructure.

TEXT BOOKS:

1.Neha Sharma, ”Green Computing for Sustainable Smart Cities: A Data Analytics Applications Perspective”,CRC Press, March 2024

2. Bhuvan Unhelkar, “Green IT Strategies and Applications-Using Environmental

Intelligence”, CRC Press, June 2014.

REFERENCES:

1. Ishfaq Ahmad, Sanjay Ranka, “Handbook of Energy-Aware and Green Computing”, Chapman & Hall , December 2020

2. Alin Gales, Michael Schaefer, Mike Ebbers, “Green Data Center: steps for the Journey”, Shroff/IBM rebook, 2011.

3. John Lamb, “The Greening of IT”, Pearson Education, 2009.